

Self-assessment questions on Diffusion

- Compare the rates of diffusion at surfaces, grain boundaries and in the bulk as a function of temperature
- Explain why interstitial diffusion is much faster than volume diffusion at same temperature
- Clear up why a diffusion mechanism is denoted thermally activated process
- State and explain the essence of the first Fick's law
- Discuss the diffusion mechanisms operating during diffusion bonding of a steel –Ti6Al4V
- Elucidate the role of diffusion during recrystallization in the TMCP of HSLA steel plates
- Compare the activation energy needed during solidification process vs. that during solid state transformation
- Detail the meaning of D_0 in the diffusion coefficient equation (hint: utilize the definition of TAP.)